

If Every Ruler Disappeared

SI units, fundamental constants and the numbers nature keeps

Opening

The metre would survive the loss of every metre rule in the world, and the kilogram would survive the loss of the platinum–iridium cylinder that once carried it, because the modern International System of Units gives certain reproducible features of nature exact numerical values and then treats those values as instructions from which any sufficiently equipped laboratory can rebuild the units.

The metre is tied to the speed of light, c ; the kilogram to the Planck constant, h ; the second to a particular hyperfine frequency in caesium-133; and the ampere, kelvin and mole to e , k_B and N_A . The SI has therefore borrowed the central idea of **natural units** while retaining metres, seconds and kilograms at sizes convenient for human life.

Two aims sit alongside one another in that arrangement. Natural units set out to reveal the universal structure of physical law by giving constants such as c and $\hbar$ the value 1, while SI units keep dimensions and conversions visible, which helps us to distinguish length from time, mass from energy, and temperature from thermal energy. Seeing why both are valuable requires us first to ask what a measurement actually is, and why constants appear in equations at all.

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1 Every measurement ends as a dimensionless number

Suppose the length of a desk is reported as

$$L = 1.73 \text{ m}.$$

The experiment has really established the ratio

$$\frac{L}{1 \text{ m}} = 1.73,$$

whose left-hand side compares two lengths, so that the dimensions cancel and the comparison yields the pure number 1.73. A measurement always compares a physical quantity with a chosen reference of the same kind.

The BIPM¹ writes this idea as

$$Q = \{Q\} [Q],$$

where Q is the physical quantity, $\{Q\}$ its numerical value and $[Q]$ its unit. Changing the unit changes the number while leaving the physical quantity untouched; a desk that is 1.73 m long is about 5.68 ft long, and neither nature nor the desk has changed in the interval.

Idea	Example	Unit-dependent?
Dimension: the physical kind of quantity	length (L), time (T), mass (M)	No
Unit: the agreed reference used for comparison	metre, second, kilogram	Yes
Numerical value: how many references fit the quantity	1.73	Yes
Dimensionless ratio: comparison of like with like	m_p/m_e	No

Table 1: The distinction that governs everything which follows.

A dimensionless number may therefore encode a genuine feature of the world.

2 Why constants enter physical laws

If two objects fall through different heights in the same gravitational environment,² the comparison can be written

$$\frac{h_1}{h_2} = \left(\frac{t_1}{t_2}\right)^2,$$

in which every ratio is dimensionless, so that the two falls can be compared without choosing metres, feet, seconds or heartbeats, and no constant appears. That comparison is local, relating

¹The Bureau International des Poids et Mesures (BIPM) is the international organisation responsible for maintaining and coordinating the SI system of measurement.

²Jean-Philippe Uzan’s review begins with this metrological story of Galileo <https://arxiv.org/html/2410.07281v2>

one fall to another performed under the same conditions. To predict a height from a time, wherever the experiment is performed, we write

$$h = \tfrac{1}{2}gt^2,$$

where the constant g turns a proportional relationship into an equality and carries the dimensions needed to make the equation balance,

$$[g] = LT^{-2}.$$

Near the Earth’s surface $g \approx 9.81 \text{ ms}^{-2}$, on the Moon it takes another value, and it therefore characterises a particular physical situation. Newton’s gravitational law makes the same move at a deeper level,

$$F \propto \frac{m_1m_2}{r^2} \qquad \longrightarrow \qquad F = G \frac{m_1m_2}{r^2},$$

and for the equality to be dimensionally consistent we require

$$[G] = L^3M^{-1}T^{-2}.$$

The numerical value 6.67430×10^{-11} belongs to G in SI units, since another coherent unit system assigns G another number; the physical strength of gravity is better expressed by a dimensionless comparison, as we shall see in §4 and §9. Units and constants together allow scientists to replace a local comparison with a reproducible global statement, so that a result obtained in one laboratory can be compared with a result obtained elsewhere and centuries later.

3 The different jobs done by constants

Calling every fixed symbol a “constant” conceals important differences, and a useful classification runs as follows.

Role	What it does	Example	Without it
Proportionality	Turns a regular relationship into a quantitative equality	g in $h = \tfrac{1}{2}gt^2$; G in Newton’s law	Trends could be compared, absolute values could not be calculated
Conversion	Translates between quantities treated separately	k_B between temperature and energy; N_A between particles and moles	Two descriptions would need separate scales
Conceptual bridge	Joins ideas once thought to be different physical kinds	c joins space and time; h joins energy and frequency	The unifying theory would be hidden
Scale or boundary	Forms a ratio marking where a new theory becomes necessary	v/c for relativity; $S/\hbar$ for quantum physics	We would lack a natural test of a theory’s domain
Unit definition	Anchors an exact reference from which units are realised	c, h, e, k_B, N_A in the revised SI	Units would depend more directly on artefacts

Table 2: Roles played by constants in physical law.

These roles overlap freely. The speed of light converts a time interval into a light-travel distance, unifies space and time, sets the relativistic speed scale and helps to define the metre; Planck’s constant converts frequency into energy, marks the scale of quantum behaviour and helps to define the kilogram; Boltzmann’s constant works largely as a conversion between kelvin and joule, while also setting the scale in $S = k_B \ln \Omega$.

Uzan follows Lévy-Leblond in offering a second classification.

Class	Meaning	Example
A	Characteristic of one particular system	the caesium transition frequency
B	Characteristic of a family of phenomena	an interaction or coupling strength
C	Universal across physical phenomena	$c, \hbar, G$, with the status of G still debated
D	A dimensionful constant selected to define units	the seven SI defining constants

Table 3: The Lévy-Leblond classification as presented by Uzan.

Class D overlaps the others, since the SI set contains universal constants, an atomic frequency, proportionality constants and the human-vision-based luminous efficacy; it was selected for stability and practical realisation, with philosophical purity a secondary consideration.

4 Constants create dimensionless tests of scale

Dimension-carrying constants become most revealing when they are used to construct dimensionless ratios.

Ratio	What its size tells us
v/c	$v/c \ll 1$: Newtonian mechanics is often sufficient; $v/c \sim 1$: relativity matters
$S/\hbar$, with S an action	$S/\hbar \sim 1$: quantum effects matter; $S/\hbar \gg 1$: classical behaviour emerges
$k_B T/(mc^2)$	$\ll 1$: thermal motion is non-relativistic; ~ 1 : thermal energies are relativistic
$GM/(Rc^2)$	$\ll 1$: weak gravity; nearer 1: strong spacetime curvature

Table 4: Dimensionless ratios as tests of a theory’s domain.

Such a constant supplies a scale against which the system can be judged. Reporting that a car travels at 30 m s^{-1} is useful engineering information, while quoting the same speed as

$$\frac{v}{c} \approx 10^{-7}$$

explains why special relativity contributes no visible correction to an ordinary journey.

5 Two species of number

Tables of dimensionful and dimensionless constants follow.

The final digits of measured quantities are revised as experiments improve, and the rounded values above express the scale students need. The numerical values of c , h and G change if our units change, whereas the values of α , μ and α_G hold fast; extraterrestrial physicists using entirely different rulers and clocks should still obtain the same dimensionless ratios, assuming the laws and constants are the same where they live.

A claimed change in a dimensionful constant is therefore ambiguous on its own, since changing the numerical value of c may amount to changing the metre or the second, whereas a change in a dimensionless ratio such as α would register as a change in observable physics.

Constant	Symbol and approximate value	SI dimensions	Main role
Speed of light	$c = 299\,792\,458\text{ m s}^{-1}$, exact	LT^{-1}	universal bridge, relativistic scale, SI definition
Planck constant	$h = 6.626\,070\,15 \times 10^{-34}\text{ J s}$, exact	ML^2T^{-1}	energy–frequency bridge, quantum scale, SI definition
Gravitational constant	$G = 6.67430(15) \times 10^{-11}\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2}$	$L^3M^{-1}T^{-2}$	gravitational proportionality and coupling
Fine-structure constant	$\alpha \approx 1/137.036$	1	electromagnetic strength
Proton–electron mass ratio	$\mu = m_p/m_e \approx 1836.15$	1	relative atomic mass scale
Gravitational structure constant	$\alpha_G = Gm_p^2/(\hbar c) \approx 5.9 \times 10^{-39}$	1	dimensionless gravitational strength between protons

Table 5: Selected entries from Tables 1 and 2 of Uzan’s review.

6 How the SI replaced objects with constants

The metre’s history shows the direction of travel, and the values below reconstruct Table 27 of Uzan’s review.

Date	Reference used for the definition	Absolute uncertainty	Relative uncertainty
1798	Earth’s meridian	0.1 mm	10^{-4}
1799	Platinum prototype	0.01 mm	10^{-5}
1889	Platinum prototype	0.1 μm	10^{-7}
1960	Krypton-86 wavelength	0.01 μm	10^{-8}
1983	Speed of light in vacuum	0.1 nm	10^{-10}
2018	Speed of light in vacuum	0.1 nm	10^{-10}

Table 6: Evolution of the metre’s realisation.

The sequence is revealing:

$$\text{planet} \longrightarrow \text{metal bar} \longrightarrow \text{atomic wavelength} \longrightarrow \text{fundamental constant}.$$

The kilogram made a similar journey. Until 2019 it depended on the International Prototype of the Kilogram, “Le Grand K”, stored near Paris, and comparisons with its official copies indicated tiny changes over time; once the numerical value of h was fixed, the cylinder changed status, ceasing to define the kilogram and becoming an object whose mass has to be measured. The definition had been moved from an ageing object to a relation built into quantum physics.

7 The seven defining constants of the SI

The table below reproduces the numerical content of Uzan’s Table 28 and the current BIPM defining-constants table, in which every listed numerical value is exact.

The last column is a convenient guide to a dependency structure, since the units are realised in sequence.

Defining constant	Symbol	Exact value	Unit fixed in	Principal anchor
Caesium-133 hyperfine transition frequency	$\Delta\nu_{\text{Cs}}$	9 192 631 770	Hz	second
Speed of light in vacuum	c	299 792 458	m s^{-1}	metre
Planck constant	h	$6.626\,070\,15 \times 10^{-34}$	J s	kilogram
Elementary charge	e	$1.602\,176\,634 \times 10^{-19}$	C	ampere
Boltzmann constant	k_{B}	$1.380\,649 \times 10^{-23}$	J K^{-1}	kelvin
Avogadro constant	N_{A}	$6.022\,140\,76 \times 10^{23}$	mol^{-1}	mole
Luminous efficacy at 540×10^{12} Hz	K_{cd}	683	lm W^{-1}	candela

Table 7: The seven defining constants of the revised SI.

1. $\Delta\nu_{\text{Cs}}$ establishes the second.
2. c , together with the second, establishes the metre.
3. h , together with the metre and second, establishes the kilogram.
4. e , together with the second, establishes the ampere.
5. k_{B} , together with the joule, establishes the kelvin.
6. N_{A} establishes the mole as an exact number of entities.
7. K_{cd} , together with the watt and a specified frequency, establishes the candela.

For the first three links,

$$1 \text{ s} = \frac{9\,192\,631\,770}{\Delta\nu_{\text{Cs}}}, \quad 1 \text{ m} = \frac{c}{299\,792\,458} (1 \text{ s}),$$

and the kilogram is the mass unit that makes

$$h = 6.626\,070\,15 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

exactly true.

The word “exact” here signifies that the numerical value has been stipulated as part of the definition, with experimental uncertainty transferred to the practical realisation of the unit and to those quantities that are no longer fixed.

8 Uncertainty was moved rather than abolished

Uzan’s Table 29 captures the conceptual reversal. The table below reformats its quoted uncertainties in parts per billion, using the values considered during the SI revision; a zero means “exact by definition”.

The revised SI chose which quantities would serve as exact anchors and transferred the uncertainty to the former artefacts and material properties, so that the direction of inference reversed: before 2019 the kilogram prototype had exactly 1 kg of mass while h was measured, and now h has an exact numerical value while the prototype’s mass is measured.

Quantity	Before / ppb	Revised SI / ppb	What changed
Speed of light, c	0	0	remained exact
Planck constant, h	10	0	became exact
Elementary charge, e	5.2	0	became exact
Boltzmann constant, k_B	370	0	became exact
Avogadro constant, N_A	10	0	became exact
International Prototype Kilogram	0	10	became a measured mass
Vacuum permeability, μ_0	0	0.23	became measured
Vacuum permittivity, ε_0	0	0.23	became measured
Water triple-point temperature	0	370	became measured
Carbon-12 molar mass	0	0.45	became measured

Table 8: Where the uncertainty went.

9 Putting dimensions back around a dimensionless constant

The fine-structure constant provides the clearest example of recovering a dimension-carrying SI constant from a dimensionless property of nature. A pure number requires a law and a set of dimension-carrying anchors before it can generate dimensions, and in this example α supplies the measured physical content while the SI-defining constants e , h and c supply the dimensional scaffold.

In SI,

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} = \frac{e^2}{2\varepsilon_0\hbar c},$$

where the dimensions cancel, leaving α dimensionless with a value of about $1/137$ that measures the strength of the electromagnetic interaction.

Since the revised SI fixes e , h and c exactly, we may solve for the vacuum permittivity,

$$\varepsilon_0 = \frac{e^2}{2\alpha\hbar c},$$

and, using $\mu_0\varepsilon_0c^2 = 1$, obtain

$$\mu_0 = \frac{2\alpha\hbar}{e^2c}.$$

Two consequences follow. First, the dimensionless constant α supplies the electromagnetic strength while the exact anchors e, h, c supply the dimensional framework, and together they generate the dimension-carrying quantities ε_0 and μ_0 . Second, ε_0 and μ_0 have lost their exactness in SI, their uncertainties descending from the measured uncertainty in α ; before 2019 μ_0 was exact by definition and e was measured, and the revised SI reversed that direction of inference.

The same pattern occurs elsewhere. From

$$G = \alpha_G \frac{\hbar c}{m_p^2},$$

the dimensionless gravitational strength α_G , combined with dimension-carrying quantities, gives the SI form of G ; and the Rydberg constant can be written approximately as

$$R_\infty = \frac{\alpha^2 m_e c}{2h},$$

where α^2 supplies a dimensionless electromagnetic factor while $m_e c/h$ supplies inverse-length dimensions, so that the result is a spectroscopic scale measured in m^{-1} .

Here lies the productive relationship between the two kinds of constant: dimensionless parameters say how strongly nature behaves, and dimension-carrying constants together with chosen units say how that behaviour is expressed on a measurement scale.

10 Why SI units are especially apt for learning and experiment

Dimensions keep the physical nouns visible

In $E = mc^2$, the factor c^2 shows exactly how mass dimensions become energy dimensions,

$$M(LT^{-1})^2 = ML^2T^{-2},$$

and the dimensions of the equation expose the relationship between the quantities, so that a student who writes $E = mc$ can detect the error before inserting any data.

Conversions remain conceptually explicit

The equations

$$E = hf, \quad E_{\text{thermal}} \sim k_B T$$

show frequency and temperature being related to energy through constants carrying the required dimensions. In thermal physics $k_B T$ is an energy scale; it is not automatically the total internal energy of a system.

The magnitudes suit laboratories and daily life

Metres, seconds, kilograms, amperes and kelvins keep ordinary quantities near manageable powers of ten, so that a classroom bench is about 1 m long rather than 10^{35} Planck lengths, and a one-second event need never be written as roughly 10^{43} Planck times.

One language crosses many branches of science

SI preserves distinctions useful in mechanics, electricity, thermodynamics, chemistry and engineering, supporting traceability from a school balance to a national standards laboratory, while dimensional analysis provides a common error-checking system across all of them.

Definitions are separated from realisations

Fixing h defines the kilogram without prescribing one permanent machine, so that a Kibble balance, a silicon-sphere experiment or some superior future technique may all realise the same definition and technology can improve without rewriting the unit. For education, the visible dimensions of SI act as the grammar of quantitative physics.

11 The case for natural units

The case for natural units begins with the observation that the numerical value

$$c = 299\,792\,458 \text{ m s}^{-1}$$

contains the history of the metre and the second, and that an alien civilisation is unlikely to arrive at that integer independently.

Natural units choose units so that selected constants take the numerical value 1. In relativistic quantum physics one commonly takes

$$c = \hbar = 1,$$

often adding $k_B = 1$, and the equations respond accordingly.

SI form	Natural-unit form	What has been absorbed into the units
$E = mc^2$	$E = m$	mass and energy share one scale through c
$E = \hbar\omega$	$E = \omega$	energy and angular frequency share one scale through $\hbar$
$E_{\text{thermal}} \sim k_B T$	$E_{\text{thermal}} \sim T$	temperature is reported on an energy scale

Table 9: Equations before and after absorption of the constants.

Setting $c = 1$ amounts to choosing as the distance unit whatever light travels in one time unit, and setting $\hbar = 1$ amounts to measuring action in units of $\hbar$, so that the constants are absorbed into the unit architecture with nothing physical removed. Dimensions that SI keeps separate then become different powers of one shared scale, and with $c = \hbar = 1$,

$$[M] = [E], \quad [L] = [T] = [E]^{-1}.$$

A length may therefore be quoted as an inverse energy while still representing a length, the conversion factors being understood rather than repeatedly written.

The advantages are genuine.

- Equations display the structure of the theory with fewer conversion factors.
- Quantities related by relativity or quantum mechanics can be expressed on a common energy scale, such as eV, MeV or GeV.
- Calculations avoid the enormous and tiny SI powers that obscure order-of-magnitude reasoning.
- The system emphasises unit-invariant, dimensionless parameters such as α and the mass ratios.

Particle physicists may accordingly quote the electron mass as 0.511 MeV, which in explicit SI language is the mass-equivalent energy $m_e c^2 = 0.511 \text{ MeV}$, or a mass of $0.511 \text{ MeV}/c^2$; with $c = 1$ the distinction no longer needs writing on every line. Natural units speak the language of a theory in which space and time, mass and energy, or temperature and energy are already connected.

12 Natural units are universal in aim, yet plural in practice

There is no unique natural-unit system, since the constants chosen depend on the physics being studied.

Stoney units	anchored on G, c, e , with the electromagnetic convention included
Natural length	$\sqrt{Ge^2/(4\pi\epsilon_0 c^4)} \approx 1.37 \times 10^{-36}$ m
Natural time	$\sqrt{Ge^2/(4\pi\epsilon_0 c^6)} \approx 4.59 \times 10^{-45}$ s
Natural mass	$\sqrt{e^2/(4\pi\epsilon_0 G)} \approx 1.85 \times 10^{-9}$ kg
Natural domain	historical gravity–electromagnetism system
Planck units	anchored on $G, c, \hbar$
Natural length	$\ell_P = \sqrt{G\hbar/c^3} \approx 1.61605 \times 10^{-35}$ m
Natural time	$t_P = \sqrt{G\hbar/c^5} \approx 5.39056 \times 10^{-44}$ s
Natural mass	$m_P = \sqrt{\hbar c/G} \approx 2.17671 \times 10^{-8}$ kg
Natural domain	quantum gravity
Atomic or “Bohr” choice	anchored on e, m_e, h , or closely related conventions
Natural length	set by atomic structure
Natural time	set by atomic dynamics
Natural mass	the electron mass scale
Natural domain	atomic physics and chemistry

Table 10: Systems discussed in Uzan’s Appendix A.3.

Introducing G into every atomic calculation is as unhelpful as measuring a classroom in Planck lengths, so physicists choose a natural system adapted to the problem, one in which the important quantities sit near 1.

The universality claim is therefore subtle. The laws and constants used to build the units may well be universal, while the choice of which constants and which remaining energy unit to adopt stays conventional; natural units supply a universal style of description, with the particular ruler still chosen by the physicist.

13 The most human constant in a supposedly natural system

The candela exposes the SI compromise especially clearly, since its defining constant K_{cd} relates physical radiant power at 540×10^{12} Hz to luminous flux as judged through the standard response of the human eye. An extraterrestrial physicist could reproduce the frequency and the radiant power, while a creature whose eyes respond most strongly to infrared would arrive at some other luminosity scale; the candela is stable and reproducible, and deliberately human.

The caesium transition holds a similarly qualified status. It is a property of one particular atom, chosen because the transition is well understood and can be reproduced with extraordinary precision, and the second remains human-sized because the exact count, 9 192 631 770 cycles, was selected to preserve the inherited second.

The revised SI is therefore a carefully engineered hybrid:

universal physics as the anchor + historical human scale as the magnitude.

14 A balanced verdict

SI and natural units answer different questions.

Question	SI answer	Natural-units answer
How can laboratories reproduce and compare measurements?	Preserve stable, human-scale units tied to exact constants	Convert back to an operational system when measurements are reported
How can students see whether an equation is physically well formed?	Keep dimensions explicit	Restore the suppressed constants when checking dimensions
How can a theory’s deep structure be written compactly?	Include all conversion factors	Choose units in which the structural constants equal 1
What numbers could alien physicists agree with us about immediately?	They could reconstruct SI once we communicated its definitions	Dimensionless constants and natural-law ratios require no shared human units
Is one system uniquely “natural”?	No; SI preserves inherited magnitudes	No; Planck, atomic and particle-physics units suit different domains

Table 11: Two systems, two sets of questions.

For a high-school physicist SI is usually the better first language, since it keeps the physical categories visible and turns dimensional analysis into a means of discernment, while natural units become illuminating once the student understands exactly what has been compressed. The mature position is able to read both: in SI,

$$E = mc^2$$

displays the bridge between mass and energy, and in natural units,

$$E = m$$

displays the unity achieved by crossing it.

Questions worth leaving with a student

1. If the numerical value of c became 1 after a change of units, what physical observation, if any, would have changed?
2. Why is a change in α more operationally meaningful than a claimed change in c alone?
3. In what sense is k_B a conversion factor, and in what sense does it still carry physical meaning?
4. What information is made visible by $E = mc^2$ and compressed by $E = m$?
5. Is a system based on $c, \hbar, G$ more “natural” for describing an atom than one based on e, m_e, h ? Explain your criterion.
6. The candela is defined partly through human visual response. Is that a flaw in a universal system, or evidence that a practical measurement system must serve its users?

Source map and notes

The conceptual spine, classifications and source tables come from Jean-Philippe Uzan, “Fundamental constants: from measurement to the universe, a window on gravitation and cosmology”,

especially §§ 2.1.1–2.1.3 and Appendix A.3–A.5; the adapted tables above correspond to selected entries from Tables 1–2 and to reconstructions of Tables 27–29.

The exact seven defining values and their interpretation were checked against the Bureau International des Poids et Mesures, “SI defining constants”, and the SI Brochure, ninth edition, version 4.01, updated in 2026.

Current recommended measured values are available from the NIST 2022 CODATA fundamental constants; for example, the current recommended values are $\alpha^{-1} = 137.035\,999\,177(21)$ and $m_p/m_e = 1836.152\,673\,426(32)$. Uzan’s tables retain the CODATA values used in the paper’s compilation, and this essay rounds such measured constants where the additional digits add no pedagogical meaning.

The historical and human-facing frame also draws on Ben Stein, “The SI: quirky by nature”, *Physics World*, August 2026, pp. 29–33, transcribed from the supplied pages.